Model 2 — pooled quantile forecaster: final results and closeout

Model 2 - daily kWh quantile forecaster - FROZEN, final test CONSUMED

Status: CLOSED as a historical experiment. No meter is production-promoted. Model 2 is frozen and is not to be tuned further against this dataset.

Conclusions

- Pooled LightGBM P50 final-test WAPE: **11.76%**
- Previous-day baseline final-test WAPE: **11.96%**
- Held-out improvement: **+0.20 percentage points**
- Nominal P10–P90 interval coverage is **materially below 80%** (70.9–72.4% observed): the intervals are under-dispersed and are **not** usable as an anomaly signal as they stand.
- **No production promotion.** The project has no formally defined absolute forecast acceptance criterion.

The pooled LightGBM improved on the previous-day baseline slightly on the held-out test, but the gain was not large enough to establish a meaningful overall advantage. Validation suggested +1.63 to +2.89pp; the held-out test delivered +0.20pp.

> **WAPE is not an accuracy percentage.** 11.76% WAPE must not be reported as "88.24% accurate". WAPE is total absolute error over total actual consumption, a ratio of sums, and it is not comparable across meters with different consumption levels.

The final test is spent

`final_test_status: CONSUMED` — 2026-05-12 through 2026-07-10, first evaluated in `model2_20260918T130515Z`.

These final-test outcomes have now been read. This window is therefore NO LONGER an untouched test set for any modified Model 2. Changing hyperparameters, features, the training population (including removing ZN.16-PB), the LightGBM parameters, the quantile calibration, the promotion rule, or adding weather or a foundation model, and then re-scoring 2026-05-12..2026-07-10, does NOT produce a held-out result. Those experiments are legitimate, but this period becomes development evidence once they are run. A genuinely new final evaluation requires unseen data later than 2026-07-10.

Historical reproduction

The strongest result in this run is not that LightGBM won. It is that a rebuilt pipeline reproduced the published baselines exactly, which is what makes Task 0B and the feature construction trustworthy.

final test, n=1,344	published §11.5	reconstructed
Previous day	11.96% / 21.58 kWh	11.96% / 21.58 kWh

final test, n=1,344	published §11.5	reconstructed
Same weekday last week	21.51% / 38.83 kWh	21.51% / 38.83 kWh

Screen coverage reproduces §11.5 exactly as well: 1,740 possible meter-days, 1,652 passing the quality screen, 1,344 with a complete feature history, 77.2%. That figure is only reproduced when both rolling features require all seven prior calendar days (`min_periods=7`), which costs 744 rows relative to a permissive rule — evidence that the original implementation used the strict rule too. Locked.

Aggregate results

window	method	n	WAPE %	MAE kWh	P10-P90 coverage %
validation_1	pooled_lgbm_p50	818	15.21	8.08	70.90
validation_1	previous_day	818	18.10	9.62	
validation_1	same_weekday_last_week	818	24.64	13.10	
validation_2	pooled_lgbm_p50	1,354	15.26	13.90	71.64
validation_2	previous_day	1,354	16.89	15.38	
validation_2	same_weekday_last_week	1,354	28.15	25.64	

Final test — scored once, after every validation artifact was written. Not used for model selection, feature selection, hyperparameter tuning or threshold selection.

window	method	n	WAPE %	MAE kWh	P10-P90 coverage %
final_test	pooled_lgbm_p50	1,344	11.76	21.22	72.40
final_test	previous_day	1,344	11.96	21.58	
final_test	same_weekday_last_week	1,344	21.51	38.83	

Quantile intervals: raw model output vs monotonic post-processing

The three quantile regressions are fitted independently and can cross. Sorting them IS reordering them, so the raw predictions are preserved under their own names and are never overwritten — see `forecast_backtest_quantiles_v2.csv` (`p10_raw` / `p50_raw` / `p90_raw`, `p10_monotonic` / `p50_monotonic` / `p90_monotonic`, `quantile_crossed`, `clipped_at_zero`).

Every WAPE, MAE, interval and coverage figure in this document uses the MONOTONIC set. Raw coverage is shown beside it only for comparison.

window	n	nominal %	coverage, monotonic %	coverage, raw %	quantile crossed %	clipped at zero %
final_test	1,344.0	80	72.40	72.25	0.89	0.15
validation_1	818.0	80	70.90	70.66	2.32	0.00
validation_2	1,354.0	80	71.64	71.64	0.59	0.00

Monotonisation is not the cause of the under-coverage: the raw intervals are just as narrow. The intervals are genuinely under-dispersed.

Note on the pooled-vs-per-meter distinction: the coverage figures above are POOLED over all rows in the window. The unweighted mean of the 29 per-meter coverages is slightly different (71.54 / 71.76 / 72.32); the pooled figures are the headline.

Per-meter validation evidence

- Statistically supported improvement (95% lower bound > 0): **14 of 29**
- Strong practical improvement (point estimate ≥ 2pp): **15 of 29**
- Not evaluable: **1** (ZN.16-PB)

These two labels are deliberately kept separate and they disagree for three meters — ZN.9-PB and ZN.5-PB clear 2pp on the point estimate but their intervals include zero, while ZN.10-PB.2 misses 2pp yet has a lower bound above zero. A fixed 2pp point-estimate gate would have admitted two meters the evidence does not support and rejected one it does.

Validation results showed meter-specific improvements, but uncertainty analysis demonstrated that a fixed 2-percentage-point threshold did not consistently distinguish statistically supported improvements. No meter was therefore promoted to production based on this experiment.

meter	qualifying windows	n	prev-day WAPE %	model WAPE %	improvement pp	95% CI pp	lower > 0	>= 2pp
ZN.1-PB.1	validation_1+validation_2	79	19.69	10.24	+9.45	[+6.82, +12.36]	yes	yes
ZN.15-PB.1	validation_1+validation_2	79	14.34	10.11	+4.24	[+3.06, +6.74]	yes	yes
ZN.19-PB.1	validation_1+validation_2	79	18.97	14.81	+4.16	[+2.16, +7.17]	yes	yes
ZN.8-PB.1	validation_1+validation_2	70	20.72	16.61	+4.12	[+2.48, +6.29]	yes	yes
ZN.10-PB.1	validation_1+validation_2	79	16.47	12.83	+3.65	[+1.27, +5.76]	yes	yes
ZN.15-PB.2	validation_1+validation_2	79	19.68	16.34	+3.34	[+2.41, +4.99]	yes	yes
ZN.11-PB.1	validation_1+validation_2	79	16.43	13.11	+3.31	[+1.52, +5.94]	yes	yes
ZN.18-PB	validation_1+validation_2	79	13.15	9.91	+3.25	[+2.05, +5.78]	yes	yes
ZN.6-PB	validation_2	41	15.49	12.41	+3.08	[+0.98, +4.61]	yes	yes
ZN.20-PB.1	validation_1+validation_2	79	24.28	21.78	+2.51	[+0.76, +4.81]	yes	yes
ZN.7-PB	validation_1+validation_2	79	14.14	11.71	+2.43	[+1.08, +4.30]	yes	yes
ZN.4-PB.1	validation_1+validation_2	79	13.67	11.27	+2.40	[+1.21, +4.27]	yes	yes

meter	qualifying windows	n	prev-day WAPE %	model WAPE %	improvement pp	95% CI pp	lower > 0	>= 2pp
ZN.3-PB.1	validation_1+validation_2	77	8.03	5.64	+2.38	[+0.89, +4.22]	yes	yes
ZN.9-PB	validation_1+validation_2	79	20.03	17.81	+2.21	[-1.14, +4.11]	no	yes
ZN.5-PB	validation_1+validation_2	52	22.81	20.74	+2.08	[-7.14, +8.48]	no	yes
ZN.20-PB.2	validation_1+validation_2	79	25.88	23.90	+1.97	[-0.95, +4.35]	no	no
ZN.10-PB.2	validation_1+validation_2	79	21.96	20.06	+1.90	[-0.29, +3.66]	yes	no
ZN.1-PB.2	validation_1+validation_2	79	13.67	12.02	+1.66	[-0.30, +2.50]	no	no
ZN.14-PB	validation_1+validation_2	69	21.08	20.20	+0.88	[-0.94, +4.30]	no	no
ZN.19-PB.2	validation_1+validation_2	62	15.52	14.96	+0.56	[-0.75, +2.52]	no	no
ZN.21-PB	validation_1+validation_2	79	16.27	15.72	+0.55	[-1.30, +1.73]	no	no
ZN.12-PB	validation_1+validation_2	65	11.61	11.15	+0.47	[-1.38, +2.32]	no	no
ZN.11-PB.2	validation_1+validation_2	79	19.44	19.23	+0.21	[-1.62, +2.35]	no	no
ZN.3-PB.2	validation_1+validation_2	79	16.31	16.12	+0.19	[-1.08, +2.23]	no	no
ZN.2-PB	validation_1+validation_2	79	14.94	14.95	-0.01	[-1.07, +1.46]	no	no
ZN.4-PB.2	validation_1+validation_2	79	31.71	31.74	-0.04	[-4.32, +2.75]	no	no
ZN.8-PB.2	validation_1+validation_2	67	27.83	28.06	-0.24	[-1.76, +1.31]	no	no
ZN.17-PB	validation_2	49	8.13	8.40	-0.27	[-1.80, +0.39]	no	no
ZN.16-PB	validation_1+validation_2	79	-	-	not evaluable	-	-	-

Improvement is previous-day WAPE minus pooled-model WAPE in percentage points; positive favours the model. Intervals come from a paired moving-block bootstrap (10,000 draws, 7-day blocks) segmented so that no block crosses the validation_1 / validation_2 boundary, because those windows are scored by separately trained models. A meter/window contributes only if that window independently satisfies both minimums (120 training and 20 evaluation observations) — which is why ZN.17-PB and ZN.6-PB carry validation_2 evidence only.

ZN.16-PB — descriptive investigation

- Task 0B calendar days: **578** (2024-12-11 to 2026-07-11)
- Quality-screen-passing days: **503**
- Passing days with `target_kwh = 0`: **488** (**97.02%**)
- Passing days non-zero: **15**, first **2024-12-27**, last **2026-07-10**
- Total validated consumption: **5,421.5 kWh** over passing days, **5,830.5 kWh** over all calendar days
- Gold cross-check: **5,830.5 kWh**, 501 zero days of 519 — agrees exactly

It did contribute training rows to every pooled model, and all of them were zeros:

window	ZN.16 train rows	of pooled total	share	all-zero targets	ZN.16 eval rows
validation_1	283	8,091	3.50%	yes	30
validation_2	313	8,909	3.51%	yes	49
final_test	362	10,263	3.53%	yes	51

From 2025-01 to 2026-05 every reading carries a NEGATIVE active_power_total_kw and a NEGATIVE power_factor_average while current_phase_1..3_a stay positive and substantial (22-80 A summed). That is a reversed-CT / polarity signature: current magnitude is measured correctly but the phase relationship is inverted, so the forward-energy register never advances and validated consumption is exactly 0.0.

The fault ends on 2026-06-26 — inside the already-spent final-test window. 2026-06 has 7,983 of 9,435 readings negative (mid-month switchover); 2026-07 has ZERO negative readings, mean active_power_total_kw +15.4 kW and mean power_factor_average +0.81. The register starts advancing and the meter records 345-435 kWh/day.

That is why ZN.16-PB is unevaluable in validation (zero denominator across all 79 eligible days) yet evaluable in the final test, where it is the one meter on which the model does clearly worse than the baseline (12.61% vs 2.06% WAPE).

Descriptive decomposition of the already-spent final test:

subset	n	model WAPE %	prev-day WAPE %	improvement pp
all 29 meters	1,344	11.76	11.96	+0.20
excluding zn16	1,293	11.75	12.05	+0.30

This is reported as a description of what already happened, **not** as a reason to remove ZN.16-PB and re-score. Nothing was retrained, excluded or reweighted. The final test is already consumed, so changing the training population and re-scoring 2026-05-12..2026-07-10 would not produce a held-out comparison.